

Java Stream flatMap Cheat Sheet

1. Flatten List

...

```
List> numberLists = List.of(  
  List.of(1, 2),  
  List.of(3, 4),  
  List.of(5, 6)  
);
```

```
List result = numberLists.stream()  
  .flatMap(list -> list.stream())  
  .toList();
```

...

2. Sentences → Words

...

```
List sentences = List.of("java stream api", "flatMap is powerful");
```

```
List words = sentences.stream()  
  .flatMap(s -> Arrays.stream(s.split(" ")))  
  .toList();
```

...

3. Employee Skills Flatten

...

```
employees.stream()  
  .flatMap(emp -> emp.skills.stream())  
  .toList();
```

...

4. Convert Words to Characters

```
...
```

```
words.stream()  
.flatMap(word -> word.chars().mapToObj(c -> String.valueOf((char) c)))  
.toList();
```

```
...
```

5. Orders → Items

```
...
```

```
orders.stream()  
.flatMap(order -> order.items.stream())  
.toList();
```

```
...
```

6. Unique Words in Paragraph

```
...
```

```
Arrays.stream(paragraph.split(" "))  
.flatMap(word -> Arrays.stream(word.split(" ")))  
.map(String::toLowerCase)  
.distinct()  
.toList();
```

```
...
```

7. Cartesian Product

```
...
```

```
list1.stream()  
.flatMap(i -> list2.stream().map(j -> i + "-" + j))  
.toList();
```

```
...
```

8. Optional Flatten

```
...
```

```
opts.stream()  
.flatMap(opt -> opt.stream())  
.toList();  
...
```

9. Map> → List

```
...  
  
map.values()  
.stream()  
.flatMap(List::stream)  
.toList();  
...
```

10. Filtering inside flatMap

```
...  
  
nested.stream()  
.flatMap(list -> list.stream().filter(n -> n % 2 == 0))  
.toList();  
...
```